Connect a signal

Digital Technologies · School computers

You will

- make code run when something happens, like a timer finishing
- connect a node's signal to a function in a script
- check the connection worked

You need

- your Godot project open, with a script on a node (see the guide Add nodes and attach a script)

Before you start

- A signal is a message a node sends when something happens: a Timer finishing, a Button being pressed, two things touching. Connecting it runs your function every time.
- This guide uses a Timer, but every node's signals connect the same way.

 Click here  Look here

1

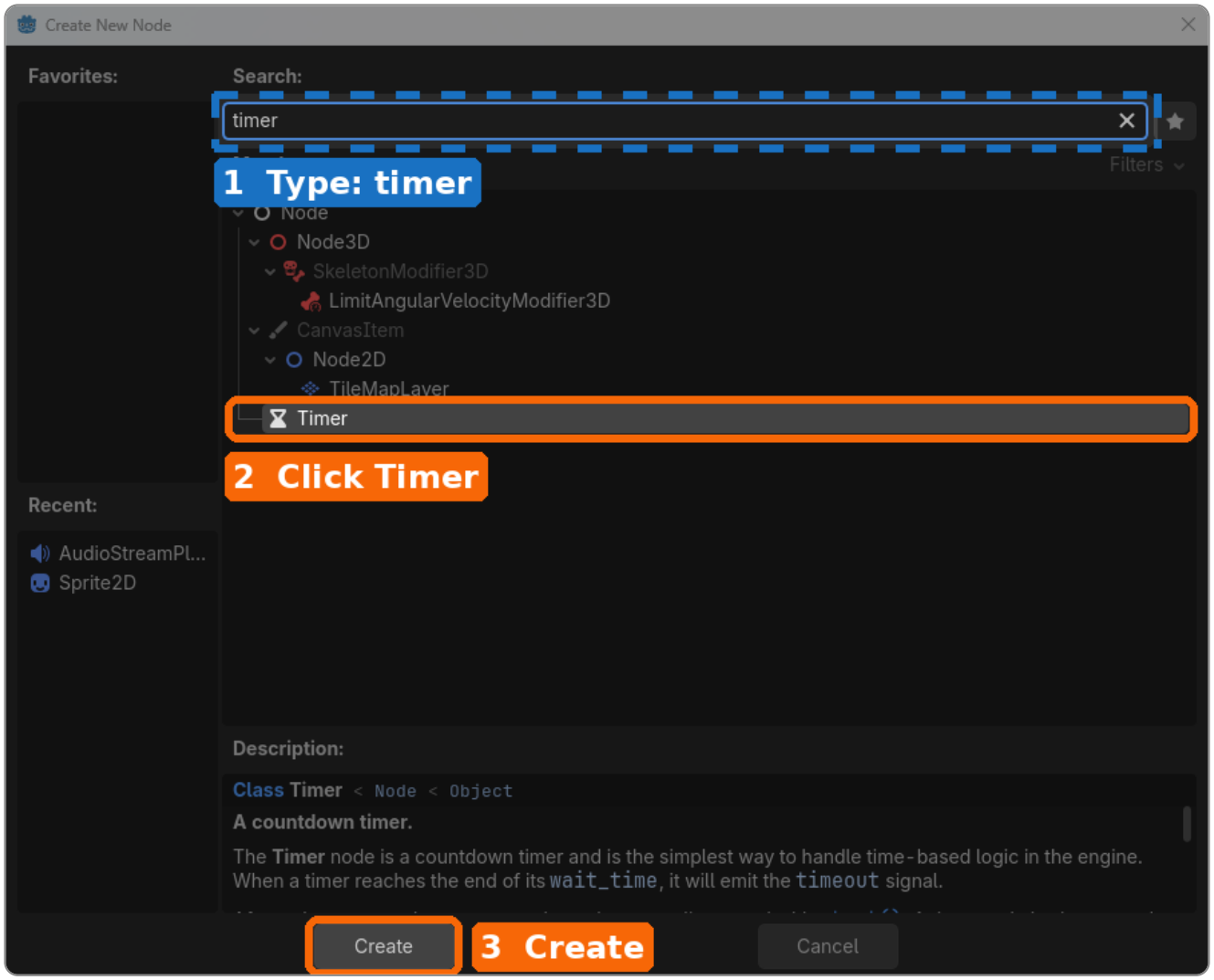
Godot › Create New Node

Click the node that has your script, then click **+** (or press `Ctrl` + `A`).

Type `timer`.

Click **Timer**, then **Create**.

The Timer goes under the node you clicked.



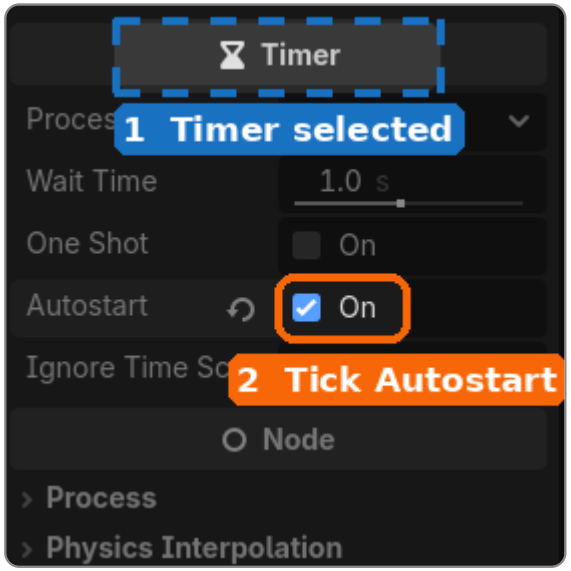
2

Godot › Inspector › Timer

With your **Timer** selected, check **Wait Time** is `1` (seconds).

Tick **Autostart**.

Autostart starts the timer when the game starts.



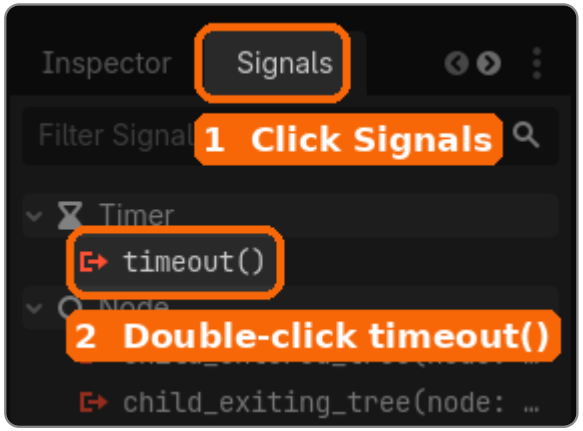
3

Godot › Signals

Click the **Signals** tab, next to **Inspector**.

Double-click `timeout()`.

The list shows every signal the selected node can send. A Timer sends timeout() each time it finishes.



4

Godot › Connect a Signal to a Method

Click the node that has your script (here, Player).

Leave the **Receiver Method** name as it is.

Click **Connect**.

The node you pick must have a script, because Godot adds the function to it.

If something goes wrong
Your node is grey: it has no script. Attach one first (see the guide Add nodes and attach a script).

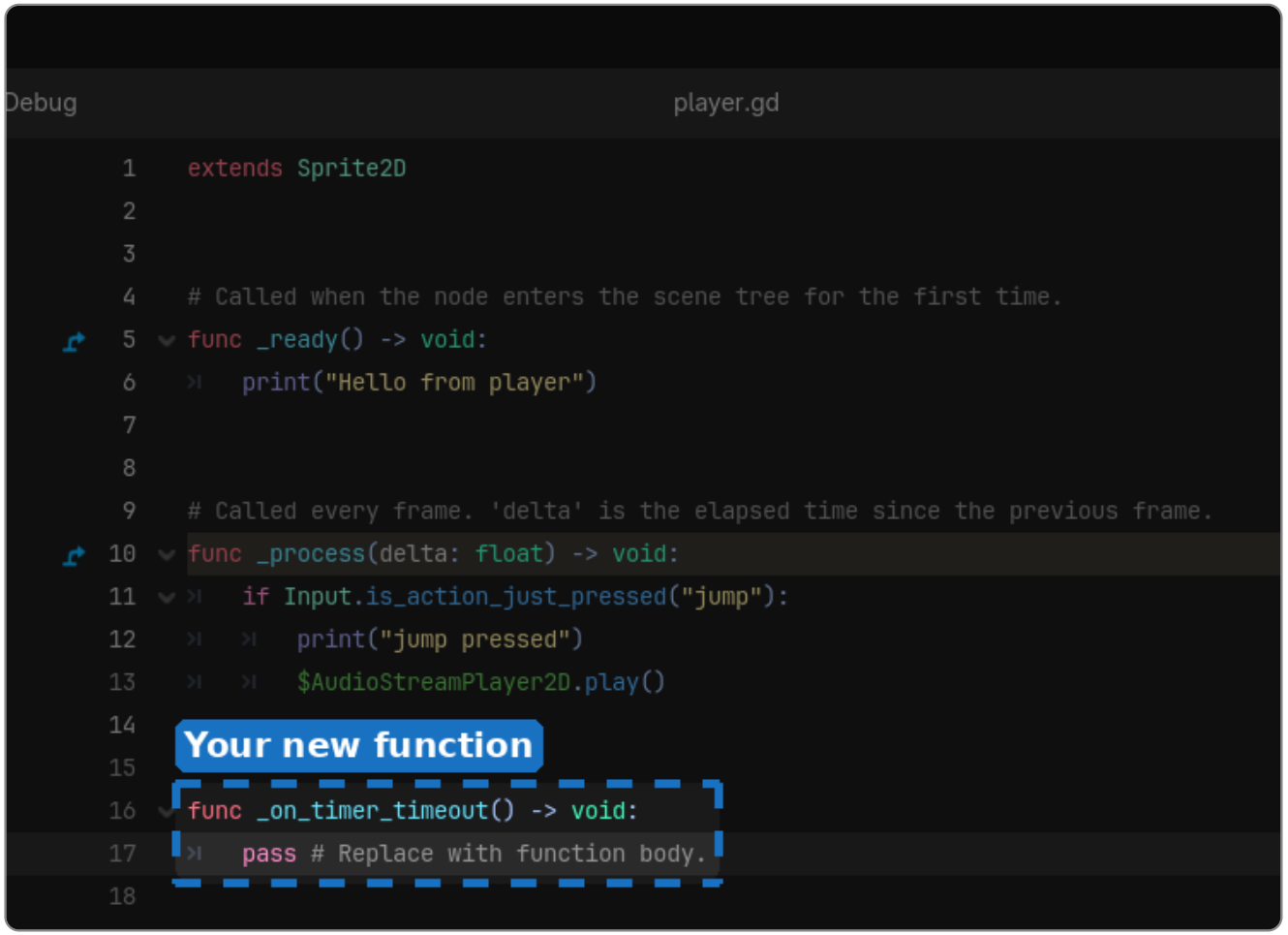


5

Godot › Script editor

Godot opens your script with a new function at the end.

The green arrow next to it shows the function is connected to a signal.



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Godot › Script editor

Replace `pass` so the function looks like this:

```
func _on_timer_timeout() -> void:
→   print("tick")
```

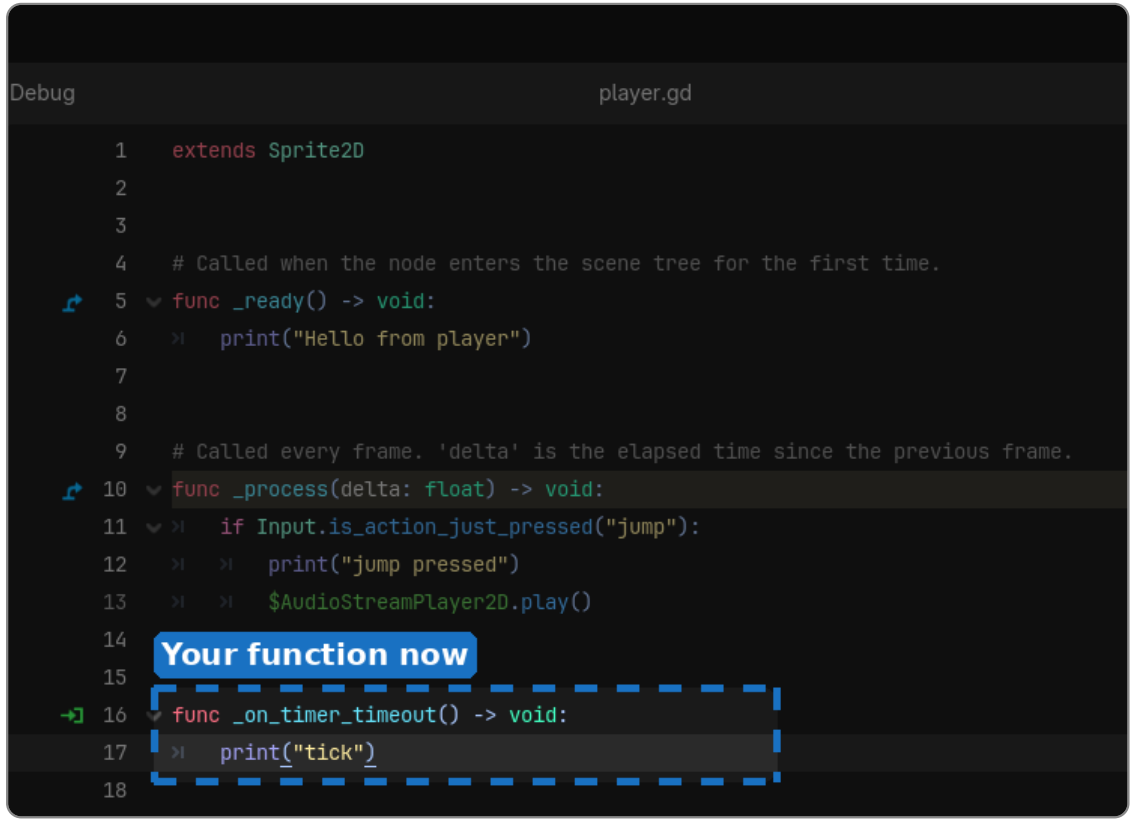
Press `Ctrl` + `S`.

Each arrow is one `Tab`.

> **Check:** the Signals tab shows your function under `timeout()`.

If something goes wrong

Error that the method doesn't exist: the function's name must match the name shown under `timeout()`, letter for letter.



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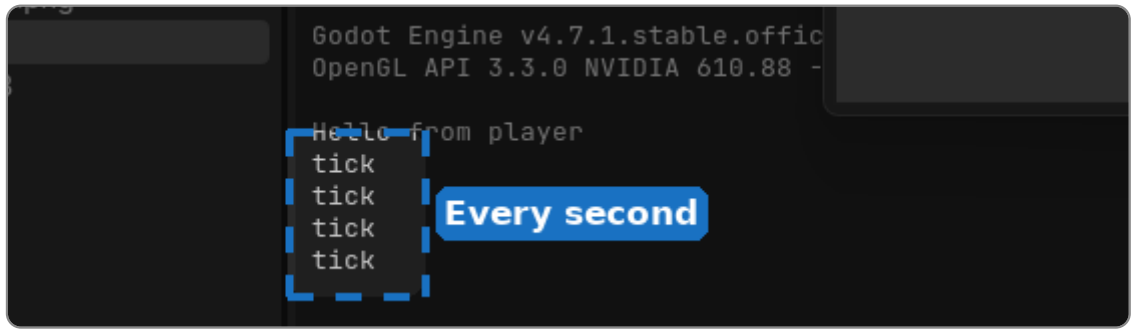
Godot › Output

Press F5 to run your game.

Watch **Output** at the bottom.

Press F8 to stop.

> **Check:** `tick` appears once every second.

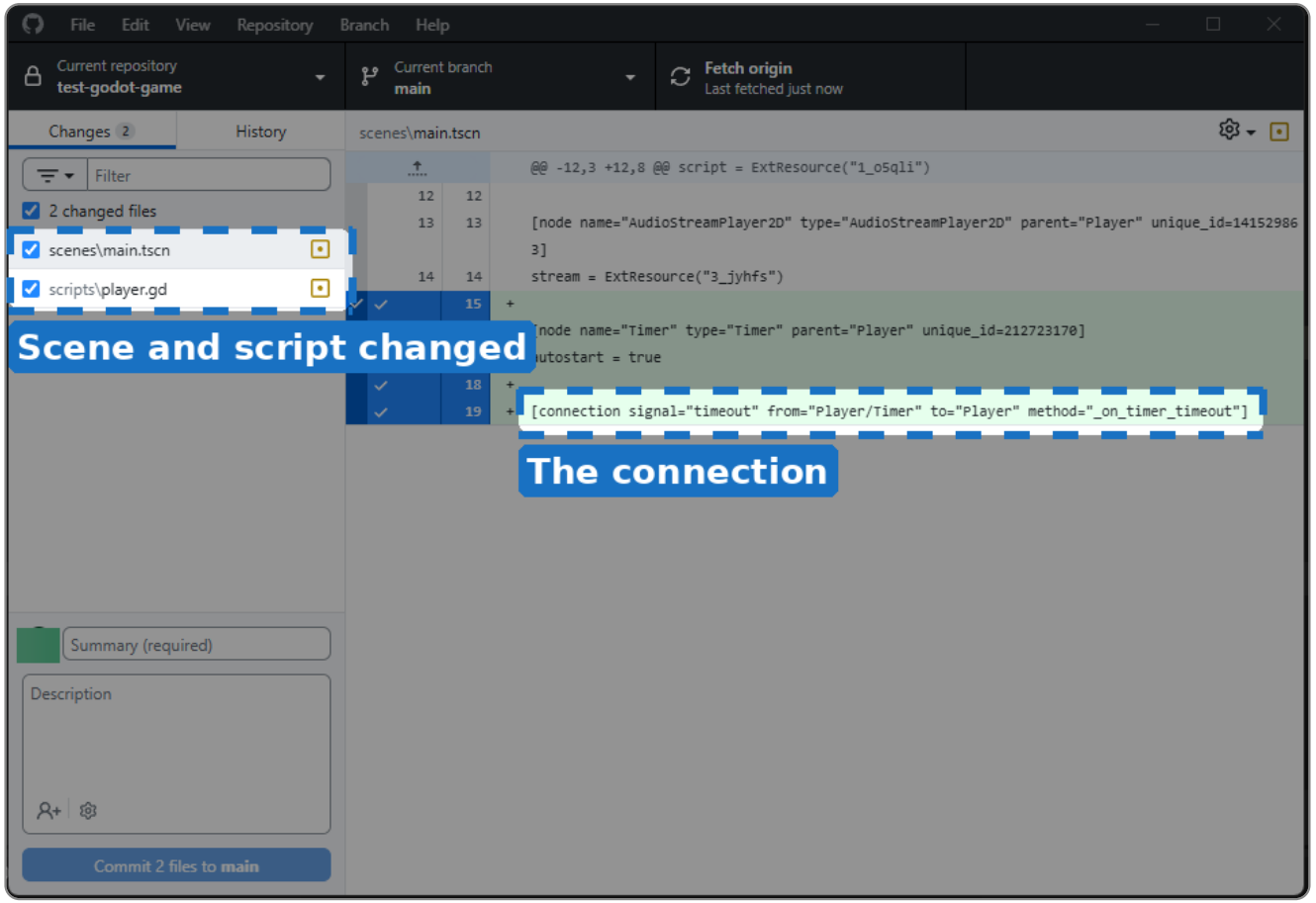


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GitHub Desktop › Changes › if your project is on GitHub

Commit and push with GitHub Desktop, as usual.

The connection is saved in your scene file, and the function in your script, so commit both.



Connecting any signal

Select the node, open the Signals tab, double-click the signal, pick the node with the script, and click Connect. Then write what should happen inside the new function.

Signals you'll use

Timer `timeout()` · Button `pressed()` · Area2D `body_entered()` · AnimationPlayer `animation_finished()`

Checkpoint

- [ ] I can connect a signal to a function.
- [ ] I can see which signals are connected in the Signals tab.
- [ ] My function runs when the signal happens.